

REVIEW

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A comprehensive review of artificial intelligence in drug design: methods, applications, and challenges

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Recently, the rapid advancement of artificial intelligence (AI), especially deep learning, has brought revolutionary changes to drug discovery, effectively addressing critical challenges of the traditional methods: high costs, extended development cycles, and low hit rates. This paper presents a comprehensive review of AI methodologies, applications, and associated challenges within drug design. Initially, the fundamental molecular representations such as strings, graphs, fingerprints and essential public databases for building AI models are introduced. Subsequently, this paper summarizes the advancements and challenges across three core subtasks in AI-based Drug Design (AIDD): (1) AI-driven virtual screening, which integrates ligand and structure-based strategies *via* discriminative and generative models; (2) AI-driven *de novo* drug design, which leverages models like generative models to navigate the vast chemical space; and (3) AI-driven drug property optimization, focusing on the multi-objective refinement of molecular attributes. Furthermore, a systematic view of the evaluation metrics is provided to benchmark different AI methods. Finally, this paper discusses the current limitations of the state-of-the-art AI methods and the potential future direction for AIDD, aiming to guide future innovations in the field.

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1 Introduction

The rapid evolution of clinical medicine towards precision and personalization necessitates the development of a new generation of therapeutics.^{1,2} Small molecule drug design has become the key research area in targeted therapy studies, distinguished by its convenient administration routes, low synthesis costs, and high stability compared to RNA and protein drugs.³

In recent years, the advancement of AI technology has brought revolutionary changes to drug discovery, effectively addressing critical challenges of the traditional methods: high costs, extended development cycles, and low hit rates.⁴ AI methodologies permeate the entire traditional drug design pipeline⁵—ranging from target discovery to hit identification, lead optimization, preclinical studies, and clinical trials—playing a particularly pivotal role in hit identification and lead optimization, as illustrated in Fig. 1. AI-driven virtual screening^{6–10} can identify small molecules that interact with biological targets from large-scale datasets by directly predicting the binding free energy or simulating the docking process and scoring binding affinities, thereby significantly reducing experimental time and costs. Meanwhile, AI-driven *de novo* molecule generation^{11–14} enables the search for potential drug candidates within the vast chemical space, transcending the

data limitations of existing molecular databases to efficiently discover novel compounds. Furthermore, AI-driven molecular property optimization^{15–18} can optimize multiple, potentially conflicting medicinal properties, such as Absorption–Distribution–Metabolism–Excretion–Toxicity (ADMET),¹⁹ Quantitative Estimate of Druglikeness (QED),²⁰ and Synthetic Accessibility score (SAscore)²¹ in parallel, assisting human experts in making informed optimization decisions.

With the continuous development of experimental techniques, the scale of molecular databases²² has expanded to the magnitude of 10⁹, with bioactive molecules²³ constituting

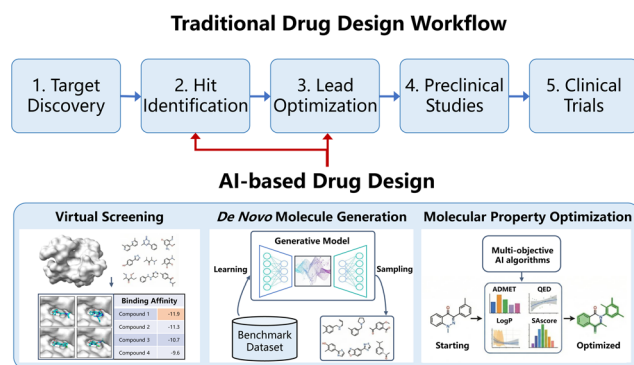


Fig. 1 Drug design workflow integrated with AI methods. The application of AI methods effectively addresses the critical challenges of the traditional drug design: high costs, extended development cycles, and low hit rates.

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a subset of approximately 10^6 . In contrast, the number of small molecule drugs approved by the US Food and Drug Administration (FDA)²⁴ is currently only around 2000. Therefore, screening for molecules within these databases that may interact with biological targets serves as an effective strategy for the identification of hit compounds. AI-driven virtual screening can significantly accelerate this process by bypassing traditional wet-lab assays in favor of direct computational predictions.⁶ Nevertheless, hampered by data sparsity and inherent algorithmic constraints, these methods frequently yield high false positive rates.²⁵

While virtual screening explores existing databases, the majority of potential drug candidates remain undiscovered within the immense chemical space, which is estimated to range from 10^{23} to 10^{60} molecules.²⁶ *De novo* molecule generation aims to generate diverse and novel molecules with druglikeness from the vast chemical space. The application of AI algorithms in this domain can effectively improve the efficiency and precision of the generation, thereby enhancing the success rate of generating valid molecules.¹¹ Conditional generation methods introduce greater controllability to *de novo* molecule generation, such as using protein binding pockets as constraints¹⁴ to generate novel interacting molecules, or utilizing specific pharmacological properties¹² as constraints to generate molecules with desired properties. However, due to the lack of robust metrics for evaluating the drug potential,²⁷ most molecules generated by AI methods cannot be directly qualified as hit compounds. In particular, the chemically implausible or atypical motifs in the generated molecules, such as complex fused ring systems, frequently draw criticism from human experts.²⁸

The transition from hit identification to lead optimization necessitates the refinement of physicochemical and pharmacokinetic profiles, quantified by metrics such as QED, SAScore, and ADMET. AI-based methods demonstrate the capability of multi-objective optimization, providing a robust solution for optimizing these interdependent and often conflicting properties in parallel.^{16,17,29}

This review summarizes the applications of AI methods in drug design, starting from the common molecular representations which serve as the input of AI models. Then the widely used public databases within the field are introduced. Furthermore, the latest research advancements and challenges across three specific subtasks: 'AI for virtual screening', 'AI for *de novo* drug design', and 'AI for drug property optimization' are analyzed. Finally, the evaluation metrics are discussed to evaluate the performance of different AIDD models.

2 Molecular representations

The primary entities of interest in AIDD are ligands and receptors. To be processed by AI models, these biological entities must be encoded into machine-readable formats, known as molecular representations. This review focuses specifically on the ligands most prevalent in small-molecule drug design—namely, small organic compounds, hereinafter referred to as 'molecules'. A single molecule can be characterized by various representation modalities, each capturing distinct structural

features from different perspectives. Consequently, selecting the appropriate representation is pivotal for enhancing model performance across diverse AIDD subtasks.³⁰ This section introduces the fundamental molecular representation methods that underpin AIDD research. As illustrated in Fig. 2, common molecular representation methods encompass string representations,^{31–33} molecular fingerprints,^{34–36} 2D and 3D molecular graphs, image-based representations, as well as other representations such as text descriptions.

2.1 String representations

String representations encode molecular structures into linear sequences of characters, facilitating efficient storage and processing, particularly for AI models adapted from Natural Language Processing (NLP).

2.1.1 Simplified Molecular Input Line Entry System (SMILES). SMILES³¹ remains the most ubiquitous string representation, encoding molecular topology using ASCII characters. It represents atoms with standard chemical symbols (*e.g.*, 'C' for carbon, 'O' for oxygen), bonds with specific characters (*e.g.*, '=' for double bond, '#' for aromaticity), and ring closures with numeric labels. For example, the SMILES string for phenylacetaldehyde is O=CCc1ccccc1. While SMILES effectively captures the topological structure information, it suffers from non-uniqueness; a single molecule yields multiple valid SMILES strings depending on the traversal order and the starting atom. Although 'Canonical SMILES' algorithms exist to standardize these outputs,³⁷ the syntax remains brittle, where minor character perturbations can result in chemically invalid structures, posing significant challenges for generative models.³⁸

2.1.2 International Chemical Identifier (InChI). InChI³² provides a distinct, standardized encoding for molecules. It features a hierarchical layered structure delimited by slashes (/), which encodes information such as chemical formula, atom connectivity, and hydrogen placement. Using phenylacetaldehyde as an example, its InChI is InChI = 1S/C8H8O/c9-7-6-8-4-2-1-3-5-8/h1-5,7H,6H2. While InChI guarantees a uniqueness mapping for each molecule, its complex syntax and rigid formatting make it difficult for generative AI models, as slight deviations typically lead to invalid representations.

2.1.3 Self-Referencing Embedded Strings (SELFIES). SELFIES³³ is a representation tailored specifically for machine learning applications, developed to surmount the validity issues associated with SMILES. It employs a self-referencing mechanism where tokens define operations (*e.g.*, ring closures and branch lengths) relative to the current state, rather than absolute syntax. For instance, phenylacetaldehyde in SELFIES is represented as [O][= C][C][C][= C][C][= C][C][= C][Ring1][= Branch1]. The defining advantage of SELFIES is its robustness; any virtual sequence of tokens corresponds to a chemically valid molecular graph. This characteristic significantly reduces the generation of invalid molecules in *de novo* generation tasks.

2.2 Molecular fingerprints

Molecular fingerprints transform complex molecular structures into fixed-length bit vectors or integer arrays, facilitating

Table 1 Classification and characteristics of common molecular fingerprints

Molecular fingerprint	Category	Characteristics
Molecular ACCess System fingerprint ³⁶ (MACCS_FP)	Substructure-based fingerprint	It is a binary vector of fixed-length (typically 166 bits), where bits denote the presence ('1') or absence ('0') of specific structural fragments
PubChem fingerprint (PubChem_FP)	Substructure-based fingerprint	It is a binary vector of fixed-length (typically 881 bits) that comprehensively covers a wide array of functional groups and pharmacophores found within the PubChem database
Extended-connectivity fingerprint ³⁴ (ECFP)	Topological fingerprint	It iteratively captures the circular topological environment of each atom up to a specific diameter. These features are mapped to a fixed-length vector <i>via</i> hashing algorithms
Topological fingerprint	Topological fingerprint	It identifies all linear subgraphs (substructures) within a defined range of bond lengths (between <i>minPath</i> and <i>maxPath</i>). These subgraphs are hashed to specific indices in the fingerprint vector
3D pharmacophore fingerprint ³⁵	3D fingerprint	It extracts spatial arrangements of pharmacophoric features (<i>e.g.</i> , hydrophobic centers, charge distributions, distances, and angles) from molecular 3D conformers and projects them onto a predefined spatial grid

undirected graph comprising 9 nodes (8 carbon nodes and 1 oxygen node) and 9 edges. The resulting adjacency matrix A has dimensions of 9×9 with binary entries ($A_{ij} \in \{0, 1\}$) denoting the existence of covalent bonds between atom pairs. Correspondingly, the node feature matrix $x_n \in \mathbb{R}^{9 \times m}$ denotes the atom type, where m represents the dimension of One-Hot encoding, while the edge feature matrix $x_e \in \mathbb{R}^{9 \times 1}$ denotes the bond types of each edge, including single, double, and aromatic bonds.

The 2D graph representation effectively captures the connectivity among constituent atoms in a molecule, providing a natural caption of a molecule while strictly preserving topological information. Generally, there exists a bijective mapping between a molecule and its 2D graph if the order of node indices is ignored. Unlike string representations or molecular fingerprints, 2D graph representations obviate the need for complex rule-based serialization and decoding schemes. Consequently, they can be directly processed by deep learning models such as Graph Neural Networks (GNNs), enabling tasks such as *de novo* molecule generation.

However, a critical challenge arises from the lack of inherent node ordering in molecular graphs, in sharp contrast to sequence-based representations.⁴⁵ This leads to the graph isomorphism problem: the same underlying molecular graph can be represented by multiple distinct adjacency and feature matrices, depending on the arbitrary permutation of node indices.⁴⁶ Failure to account for this invariance can result in model instability, reduced diversity in generated structures, and compromised the prediction accuracy especially in graph-based generative and predictive tasks.

2.4 3D molecular graphs

Similar to 2D graph representations, 3D graph representations also model molecules as graph $G = (V, E)$. However, a key

distinction lies in the definition of edges: in 3D molecular graphs, edges are not governed solely by covalent bonds. In 3D space, interatomic interactions such as van der Waals forces or electrostatic interactions exist even between non-bonded atoms. Consequently, edge construction in 3D graphs typically relies on spatial proximity rather than just covalent connectivity.⁴⁷ One common approach involves setting a distance cutoff threshold r , where an edge is established if the Euclidean distance between two atoms falls below this threshold. Alternatively, a fully connected graph structure can be employed, where edges exist between all atom pairs, with the Euclidean distances serving as edge weights. Similar to 2D molecular graph, a 3D molecular graph is formalized using an adjacency matrix $A \in \mathbb{R}^{|V| \times |V|}$, a node feature matrix $x_n \in \mathbb{R}^{|V| \times (m+3)}$, and an edge feature matrix $x_e \in \mathbb{R}^{|E| \times n}$. The defining characteristic of the 3D graph is that the node feature matrix explicitly incorporates 3D atomic coordinates.

Distinct from traditional 3D descriptors (*e.g.*, 3D pharmacophore fingerprints) that often compress spatial information, 3D graph representations preserve exact atom coordinates. Consequently, they serve as the fundamental substrate for geometric deep learning models. By operating directly on raw molecular coordinates, these models retain strict geometric fidelity. Since the 3D conformation of a molecule dictates its biological function, 3D graph representations are extensively applied in tasks such as protein–ligand interaction prediction. Furthermore, due to the dynamics of small molecules and their conformational changes during interactions with other molecules *in vivo*, an increasing number of methods are focusing on the molecular conformational variations rather than a single energy-minimized conformation.

In relative tasks involving 3D molecule generation or interaction prediction, addressing geometric equivariance is critical.⁴⁸ This refers to the physical principle that the properties of a molecule like potential energy remain invariant under

geometric transformations of its coordinates, such as rotation, translation, and reflection. To address this, recent research has introduced Equivariant Graph Neural Networks (EGNNs)⁴⁹ designed to preserve these symmetries. Alternatively, geometric transformations can be applied as data augmentation methods during training. Both strategies have proven effective in enhancing the model accuracy and generalization capabilities.

2.5 Molecular images

Propelled by rapid advancements in Computer Vision (CV) algorithms, AI models have achieved exceptional performance in image classification and generation tasks.^{50,51} Notably, modern generative models have reached remarkable success in capturing complicated local details while retaining global image coherence, which satisfies the requirements of molecular design. Molecular image representation transforms molecular structures into 2D grids⁵² or 3D voxels. These inputs are then processed *via* the visual perception capabilities of AI models to execute tasks such as molecular design and property classification.

However, the application of image-based representations within the drug design domain remains relatively sparse.^{53,54} The existing body of research concentrates on *de novo* design and structural optimization, with most studies utilizing gray-scale images of 2D molecular structures.

2.6 Other representations

Less conventional representation methods, such as textual descriptions,^{55,56} characterize molecules using nomenclature, functional annotations, substructures, and associated targets. However, as this approach relies heavily on information from existing databases, corresponding descriptive text is available for only a limited subset of molecules.⁵⁷

Despite the availability of diverse molecular representations, the selection of the optimal approach or combination thereof must be tailored to specific tasks and datasets. Table 2 summarizes the respective advantages, limitations, and

application scenarios in AIDD for the five aforementioned representations. A deep understanding and appropriate application of these representations are important for advancing research across various subtasks in AIDD.

3 Benchmark datasets

AI algorithms, particularly deep learning models, are fundamentally data-driven. Their core mechanism involves learning underlying data distributions or approximating the mapping functions between inputs and target outputs. Complex deep learning architectures often comprise millions of parameters, the training and fine tuning of which rely heavily on large-scale datasets.⁵⁸ Fortunately, drug design benefits from a rich repository of databases that deposit diverse molecular data modalities, ranging from 2D molecular structures and 3D conformations to knowledge graphs, bioactivity profiles, and textual descriptions. These resources serve as the essential data infrastructure for training AI models. This section provides an overview of some public databases and datasets widely used in drug design tasks, as summarized in Table 3, followed by an in-depth introduction of several databases.

3.1 Quantum mechanics results for organic molecules with up to 9 heavy atoms (QM9)

The QM9 (ref. 44) dataset contains 133 885 stable organic molecules with up to 9 heavy atoms drawn from the atom set [C, O, N, F]. For each entry, the dataset provides geometric and physicochemical properties, including the lowest energy conformation, harmonic frequencies, dipole moments, polarizability, and various thermodynamic parameters, all derived *via* Density Functional Theory (DFT) calculations.

While QM9 is extensively employed as a benchmark set for *de novo* 3D molecular design and property prediction, it presents inherent limitations: (1) the dataset is restricted to very small molecules, covering only a narrow fraction of the relevant chemical space; (2) it provides only a single static lowest-energy

Table 2 Comparison of different molecular descriptors

Descriptors	Advantages	Limitations	Tasks
Strings	Compact storage and ease of text-based editing and processing	Non-uniqueness; requires strict syntax rules to ensure chemical validity	<i>De novo</i> generation, virtual screening, property optimization
Fingerprints	Fixed-length vectors; computationally efficient for rapid similarity calculations	Irreversible information loss; inability to reconstruct original molecular structures	Molecular similarity search, virtual screening
2D graphs	Explicitly captures 2D topological connectivity and atom-bond relationships	Lacks 3D conformational information; sparsity of adjacency matrices; graph isomorphism (matching) challenges	<i>De novo</i> generation, virtual screening, property optimization
3D graphs	Incorporates comprehensive 3D spatial and conformational information	Dynamic nature of conformations; high computational cost; complexities regarding geometric equivariance	<i>De novo</i> generation, virtual screening, property optimization
Images	Intuitive visualization; high interpretability	Difficulty in molecular reconstruction; challenges in model training	<i>De novo</i> generation, property optimization

Table 3 Common molecular databases in AIDD and their applications

Database	Data type	Description	Scale	Applications	URL
QM9 (ref. 44)	3D	3D conformations & quantum properties	1.3×10^5	<i>De novo</i> generation, property optimization	https://dx.doi.org/10.6084/m9.figshare.978904
GEOM ⁵⁹	3D	3D conformations	3.7×10^7	<i>De novo</i> generation, property optimization	https://doi.org/10.7910/DVN/JNGTDF
PDBbind ⁶⁰	3D	Protein–ligand complex structures	2.7×10^4	<i>De novo</i> generation, target-based generation, virtual screening	https://www.pdbbind-plus.org.cn/
ChEMBL ²³	2D	Bioactive drug-like molecules	2.5×10^6	<i>De novo</i> generation, virtual screening	https://www.ebi.ac.uk/chembl/
ZINC ²²	2D	Commercially available compounds	9.8×10^8	<i>De novo</i> generation, virtual screening	https://zinc15.docking.org/
MOSES ⁶¹	2D	High-quality molecule subset	1.7×10^6	<i>De novo</i> generation, virtual screening	https://github.com/molecularsets/moses
PubChem ⁶²	2D/3D	Chemical structures & bioassays	1.2×10^8	<i>De novo</i> generation, virtual screening, property optimization	https://pubchem.ncbi.nlm.nih.gov/
DRKG ⁶³	KG ^a	Drug-disease relations	9.7×10^4	Conditional generation, virtual screening	https://github.com/gnn4dr/DRKG
CheBI ⁵⁷	Text	Text descriptions of compounds	10^6	Conditional generation, virtual screening	https://www.ebi.ac.uk/chebi/

^a 'KG' represents 'Knowledge Graph'.

conformation for each molecule, limiting its applicability in modeling molecular dynamics or flexible conformational changes.

3.2 Geometric Ensemble of Molecules (GEOM)

GEOM⁵⁹ dataset comprises a vast repository of over 37 million molecular conformations corresponding to more than 450 000 molecules. Notably, GEOM serves as a superset of QM9, containing all its entries while extending to significantly larger chemical systems. The molecules in GEOM feature a maximum size of 181 atoms (91 heavy atoms) and an average of 44.4 atoms, representing a substantial increase in structural complexity compared to QM9.

These data were calculated using a combination of molecular dynamics (MD) simulations for enhanced sampling and semi-empirical DFT. By effectively covering medium-sized, drug-like chemical space, GEOM offers superior applicability for drug design tasks. An advantage of GEOM is its provision of multiple low-energy conformations for a single molecule, which facilitates the training of AI models capable of capturing structural dynamics. Current AI methods frequently employ subsets of this data, such as the top 30 lowest-energy conformations per molecule.¹²

3.3 ChEMBL

ChEMBL²³ stands as a manually curated database of bioactive molecules with drug-like properties. It aggregates bioactivity data extracted from primary scientific literature (spanning from 1974 to the present), encompassing approximately 2.5 million distinct compound records and associated activities against nearly 16 000 biological targets.

As a definitive resource of bioactivity information, ChEMBL is extensively employed in tasks such as ligand-based virtual screening, target prediction, and *de novo* molecule generation.

4 AI for virtual screening

Despite rapid advancements in High-Throughput Screening (HTS)⁶⁴ and synthetic chemistry, the simultaneous screening of massive compound databases remains a challenging task. To address this problem, computational chemists introduced virtual screening which utilizes computational models and simulation methods to identify potential ligand from vast chemical databases that interact with protein targets, thereby controlling costs and saving time.⁶⁵ Over decades of development, virtual screening methodologies have transitioned from traditional physics-based energy models to data-driven artificial intelligence frameworks.⁶⁶

Traditional virtual screening strategies are classified into structure-based (SBVS) and ligand-based (LBVS) approaches, contingent upon the availability of the structure data of protein targets.⁶⁷ When the 3D structure of the target protein is undetermined, LBVS⁶⁸ serves as the primary strategy. It utilizes algorithms such as QSAR, similarity analysis, and pharmacophore mapping to identify similar structures of known active compounds from large databases. In contrast, once the target structure and binding pocket are resolved, SBVS⁶⁹ methods like molecular docking are utilized to predict binding affinity through the calculation of ligand–protein binding free energies. To optimize the screening efficiency, these methodologies are frequently combined in a sequential workflow: LBVS is first applied to rapidly narrow the candidate space, after which SBVS provides a more detailed assessment of binding affinity, as illustrated in Fig. 3.

Correspondingly, the application of AI algorithms in virtual screening generally follows two distinct paradigms: (1) discriminative models,^{8,9,25,70–75} which function as predictors to determine the probability of interaction or quantify the binding affinity for a given ligand–protein pair, hereinafter referred to as

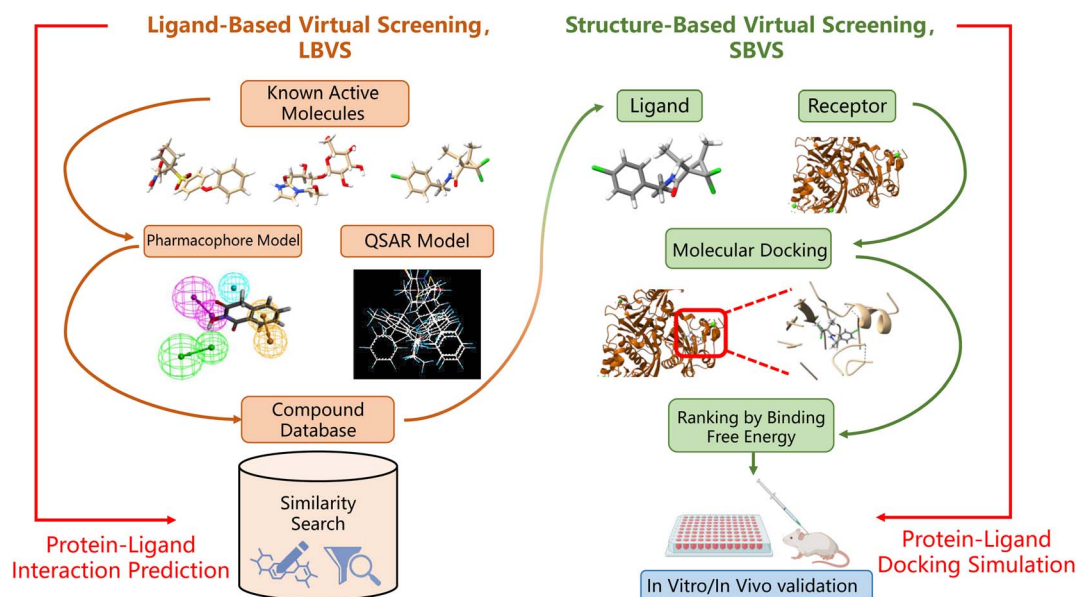


Fig. 3 Overview of LBVS and SBVS workflows and the applications of AI-based methods. LBVS identifies the similar structures with the known active molecules from databases, while SBVS assess the binding affinities of the ligands by simulating the docking structures with the protein target.

‘Protein–Ligand Interaction Prediction (PLIP)’ and (2) generative models,^{6,7,10,43,76–80} which are employed to model the interaction landscape, aiming to explore optimal binding conformations and identify key interaction sites, hereinafter referred to as ‘Protein–Ligand Docking Simulation (PLDS)’.

PLIP approaches first extract diverse features or employ molecular representations to encode both proteins and ligands (or just ligands). These representations serve as inputs for supervised AI models, which are trained to perform classification or regression tasks using interaction status or quantitative affinity metrics such as the dissociation constant (k_d), inhibition constant (k_i), and Michaelis constant (k_m) as ground truth labels.

However, these feature-based PLIP methods face significant challenges stemming from the inherent limitations of current databases, specifically data sparsity, insufficiency, and class imbalance. A protein target typically interacts with only a restricted group of ligands, resulting in a long-tail distribution where positive samples are vastly outnumbered by negatives.⁸¹ Furthermore, given the prohibitive cost of high-throughput biological assays, the list of interactive ligands for most proteins remains incomplete. Consequently, unlabeled protein–ligand pairs cannot be naively categorized as negative samples, complicating the training process.

Notably, this imbalance encourages deep learning models to leverage topological shortcuts instead of learning generalizable patterns from the training data.⁸² Recent research studies²⁵ reveal that multiple state-of-the-art models tend to predict interactions based on the degree information within the protein–ligand bipartite network rather than learning binding patterns from the molecular structures. While this phenomenon helps AI models yield high performance on some datasets, it fails to generalize to novel, unseen molecules. To address this,

strategies such as unsupervised pre-training and data augmentation have been introduced to PLIP to mitigate the imbalance and improve the generalizability.^{25,73}

Graph-theoretic approaches⁸³ offer a different strategy by modeling interactions as edges within a heterogeneous protein–ligand network, casting the problem as link prediction. However, these methods face severe challenges in the inductive setting, specifically, processing novel entities undefined in the current protein–ligand network. While training a mapping networks can approximate embeddings for new entities *via* similarity alignment, this often introduces additional errors and raises concerns regarding information leakage.

Table 4 summarizes recent AI-based algorithms for PLIP. Among these methodologies, GNNs and their variants have emerged as the dominant paradigm, driven by several advantages. Primarily, as discussed in Section 2, the molecular graph offers a natural topological representation, aligning more closely with the physicochemical nature of molecules than sequence-based or image-based representations. Additionally, regarding 3D structural modeling, the binding affinity is thermodynamically governed by non-covalent interactions such as van der Waals forces, electrostatics, and hydrogen bonds, while these interactions can be modeled *via* geometric features such as interatomic distances and angles in 3D graph representations. Furthermore, given that protein–ligand interactions are spatially localized to specific binding pockets, the GNN architecture is suited to discover critical binding residues owing to its ability to aggregate features from local to global scales. Moreover, compared to dense architectures like CNNs, GNNs typically exhibit parameter efficiency, facilitating faster convergence on the relatively sparse datasets in PLIP.

LigBind⁷² introduces a transfer learning framework, developed to mitigate challenges such as the long-tail distribution of

Table 4 Recent AI-based algorithms for protein–ligand interaction prediction

Method	Model	Year	Reference
DeepPocket	CNN	2021	70
GraphBind	GNN	2021	8
ScanNet	Attention, GNN	2022	71
LigBind	GNN	2023	72
NABind	Template, GNN	2023	84
AI-Bind	Network-based	2023	25
ZeroBind	GNN	2023	73
PIGNet2	GNN	2024	74
HybridGeo	GNN	2025	75
CORDIAL	Attention, CNN	2025	9

the data and the complexity of ligand binding modes. By leveraging the underlying similarity in binding modes across diverse ligands, the method employs a pre-training and fine-tuning paradigm to improve the classification accuracy and robustness on imbalanced datasets. Ablation studies confirm that this transfer learning strategy effectively enhances the model performance; notably, LigBind significantly outperforms baseline methods on unseen ligands, demonstrating superior generalization capabilities.

HybridGeo⁷⁵ presents a geometric deep learning framework designed to address the limitations of the homogeneous representation of atomic interactions and the underutilization of spatial information. By leveraging dual-view geometric graphs to differentiate intra- and inter-molecular interactions, the method employs a novel hybrid message-passing strategy that utilizes spatial distances as both messages and weights. This approach captures 3D geometric features and residue dependencies. The evaluation results demonstrate that HybridGeo achieves state-of-the-art performance on the PDBbind benchmark dataset and exhibits generalizability on the macrocyclic compound complexes.

CORDIAL⁹ proposes a deep learning framework designed to overcome the generalizability limitations of current structure-based affinity prediction models which often learn spurious structural correlations. By employing an interaction-centric embedding strategy, the method constructs distance-dependent interaction radial distribution functions to explicitly encode physicochemical rules while avoiding direct parameterization of specific chemical substructures. Through a specialized architecture integrating 1D feature-specific convolutions and axial attention, CORDIAL effectively captures both local and global interaction patterns. Rigorous validation using a stringent Leave-Superfamily-Out (LSO) benchmark confirms that this approach exhibits robust performance on out-of-distribution targets; notably, CORDIAL significantly outperforms other baselines in ordinal ranking tasks for novel protein families, demonstrating superior generalizability and physical interpretability.

PLDS approaches primarily utilize generative frameworks or regression-based methods to model the 3D conformation of the protein–ligand complex. These models are trained to perform conformation sampling or direct coordinate prediction,

utilizing experimentally determined crystal structures or calculated Minimum Free Energy (MFE) conformations as ground truth to minimize geometric deviations, typically quantified by the Root Mean Square Deviation (RMSD) between the generated and native structures.

However, current PLDS methodologies suffers from challenges arising from the dynamic nature of biological molecules and the complexity of the 3D conformational space.⁷⁹ Proteins often undergo significant conformational changes upon ligand binding, yet many methods treat the protein pocket as a rigid body to reduce computational complexity. The structural states of a protein in its ligand-free and ligand-bound forms are termed the apo-structure and holo-structure, respectively. Furthermore, high-quality 3D structural data are scarce compared to 1D/2D data, and the vast degrees of freedom involved in ligand torsion and protein side-chain movements create an immense conformational space. Additionally, for novel protein–ligand pairs, the specific binding pocket often remains unknown. This uncertainty significantly increases the complexity of predicting the resulting holo-structure which is termed blind docking.⁸⁵ Consequently, accurately predicting the binding pose for protein–ligand complexes while accounting for protein flexibility remains a computationally expensive and challenging task.

Table 5 summarizes recent AI-based algorithms for PLDS. The majority of these methodologies are grounded in autoregressive generative models like transformers or diffusion models to generate the 3D conformations of the binding pockets and ligands. By learning the transformation mapping from a noise prior to the data distribution, diffusion models effectively simulate the dynamic process of molecular docking.

TankBind⁷⁶ introduces a two-stage deep learning framework for protein–ligand binding structure prediction, designed to overcome the limitations of traditional scoring functions and the ignores of geometric constraints in existing graph-based methods. The method jointly infuses trigonometric constraints and excluded-volume effects as inductive biases as well as segments the protein into functional blocks to detect all possible binding sites. By applying a contrastive loss function to ensure the native block binds stronger to the compound than decoys, TankBind achieves substantial performance compared to traditional computational methods.

Table 5 Recent AI-based algorithms for protein–ligand docking simulation

Method	Model	Year	Reference
Tankbind	GNN	2022	76
DiffDock	Diffusion	2023	6
KarmaDock	GNN, MDN (mixture density network)	2023	77
SurfDock	Transformer, diffusion	2023	7
CarsiDock	Transformer, MDN	2024	78
AlphaFold 3	Transformer, diffusion	2024	10
RoseTTAFold-all-atom	Transformer	2024	43
DynamicBind	Diffusion	2024	79
FlexSBDD	Flow matching	2024	80

DiffDock⁶ presents a diffusion model for PLDS, developed to address the limitations of regression-based approaches like Tankbind. Regression-based approaches aim to select a single conformation learnt by minimizing the loss, while DiffDock expand this to some significant modes. By reformulating docking as a generative modeling task on the non-Euclidean manifold of ligand poses, this method maps translational, rotational, and torsional degrees of freedom onto a product space of groups, thereby facilitating an efficient diffusion process that iteratively refines random poses. The evaluation results confirm that DiffDock yields substantial improvements in the accuracy and robustness; notably, it exhibits superior precision when applied to PLDS on computationally structures generated by ESMFold.

However, as existing methods frequently employ coarse-grained representations to model protein–ligand interactions to reduce the complexity, this strategy may oversimplify the protein structures, which can potentially lead to intermolecular clashes. To solve this problem, SurfDock presents a surface-informed geometric diffusion model which integrates a multi-modal protein representation that fuses sequence embeddings, residue graphs, and molecular surface features.

AlphaFold 3 (AF3)¹⁰ pioneers the structural prediction of for how proteins interact with other molecules including proteins, nucleic acids, small-molecule ligands, ions, modified residues, and *et al.* Advancing beyond the architecture of AlphaFold 2 with refined modules for template matching and Multiple-Sequence Alignment (MSA) processing, AF3 incorporates a deep diffusion model to directly generate 3D atomic coordinates, outperforming traditional virtual screening methodologies.

However, its performance specifically for the structural modeling of the protein–ligand complex remains suboptimal. This limitation potentially stems from its design as a general framework; the model is optimized to generalize across a diverse spectrum of bio-molecules, rather than being exclusively tailored to protein–ligand binding task.

5 AI for *de novo* drug design

In contrast to virtual screening, which is limited by the current molecular databases, *de novo* generation methods extend the search for novel structures to the vast chemical space which encompassing all theoretically feasible small organic molecules. The integration of AI methods, particularly deep generative models, has established a new paradigm in *de novo* drug design, significantly improving the generation efficiency and the model performance on metrics such as novelty and diversity.

AI-based *de novo* drug design can be categorized into molecule generation and Structure-Based Drug Design (SBDD). Molecule generation aims to produce a cohort of molecules characterized by chemical validity, novelty, and diversity, while SBDD distinguishes itself from this unconstrained generation by introducing specific conditional constraints, namely the incorporation of target protein structures or binding pocket information to guide the generation process.

Generative AI models were introduced to molecular generation around 2017. These approaches encompass a diverse range of architectures, including Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Normalizing Flows (Flows), Recurrent Neural Networks (RNNs), and diffusion models, among others, as summarized in Table 6.

VAE is the pioneering deep learning architecture applied to *de novo* molecular generation. Early approaches, such as the Chemical VAE (CVAE),⁸⁶ leverage the VAE framework to map discrete SMILES sequences into a continuous latent space. Within this latent space, operations such as random sampling, interpolation, and gradient-based optimization are performed to explore and decode novel molecules.

Despite achieving the automated exploration of chemical space, sequence-based VAEs suffer from some limitations: (1) constrained exploration space: as the training objective minimizes the distributional divergence between generated and training samples, the learned latent space is bounded by the training distribution. Since the training sets represent only a narrow fraction of the chemical space, the generated molecules frequently exhibit limited novelty and diversity. (2) Syntactic fragility: SMILES strings are governed by rigid syntax rules; consequently, even minor perturbations during encoding or decoding can result in syntax errors, leading to the generation of chemically invalid molecules. Moreover, the SMILES representation lacks smoothness, meaning that small changes in the latent space can correspond to large structural variations.

To address these issues, the Junction Tree VAE (JT-VAE)³⁸ was introduced as a pioneering graph-based framework. JT-VAE decomposes molecules into chemically valid substructures (*e.g.*, rings and bonds) to construct a junction tree representation. It employs a novel dual-channel architecture: a graph encoder learns atomic-level features, while a tree encoder captures topological connectivity, effectively decomposing the latent representation into tree scaffolds and assembly rules. During the generation, a tree decoder predicts substructures *via* Depth-First Search (DFS) traversal, while a graph decoder

Table 6 Recent AI-based algorithms for molecule generation

Method	Model	Representation	Year	Reference
CVAE	VAE	SMILES	2018	86
GVAE	VAE	SMILES	2017	93
SD-VAE	VAE	SMILES	2018	94
JT-VAE	VAE	2D graph	2018	38
druGAN	GAN	Fingerprint	2017	87
MolGAN	GAN	2D graph	2018	88
Graphvae	VAE	2D graph	2018	11
GraphNVP	Flow	2D graph	2019	95
Moflow	Flow	2D graph	2020	90
GraphAF	Flow	2D graph	2020	89
BIMODAL	RNN	SMILES	2020	91
EDM	Diffusion	3D graph	2022	12
GeoLDM	Diffusion	3D graph	2023	92
MUDiff	Diffusion	2D graph + 3D graph	2024	96
SketchMol	Diffusion	Image	2025	53

assembles them. By systematically enumerating and scoring connection topologies, JT-VAE guarantees chemical validity by design, achieving 100% validity without reliance on post-processing methods.

GANs are characterized by a competitive dynamic between two neural networks: a generator and a discriminator. The generator is tasked with mapping random noise to generated samples that mimic the training data, while the discriminator functions as a binary classifier, striving to distinguish between the positive samples from the training set and the negative samples produced by the generator. This process is formalized as a minimax two-player game, driving the model to converge when the generated data distribution approaches the true data distribution.

GAN models have raised significant attention in molecular generation. For instance, druGAN⁸⁷ employs a dynamic training strategy to substantially enhance the stability and accuracy of molecular fingerprint generation. Similarly, MolGAN⁸⁸ integrates GANs with Reinforcement Learning (RL) to generate molecular graph structures constrained by a fixed heavy-atom count. While these GAN-based methodologies exhibit high computational efficiency and low reconstruction loss, they often face challenges regarding to mode collapse—a phenomenon where the generator produces a limited variety of samples. Consequently, the diversity and novelty of molecules generated by GANs often lag behind those generated by VAE-based models.

Normalizing Flows constitute a class of generative models that utilize a sequence of invertible, differentiable transformations (bijections) to map simple base distributions such as Gaussian distribution to complex data distributions. In contrast to VAEs, which rely on approximate inference with distinct encoders and decoders, flow models define the generative process as the exact mathematical inverse of the inference process. This bijective nature allows for tractable likelihood computation and exact data reconstruction.

Flow-based architectures have found extensive application in molecular generation. For instance, GraphAF⁸⁹ introduces a hybrid framework that integrates autoregressive modeling with flow-based density estimation. By combining the sequential generation capabilities of autoregressive models with the exact likelihood optimization of flows, GraphAF achieves high-precision molecular graph generation with superior chemical validity, further optimize pharmacochemical properties based on RL. Similarly, MoFlow⁹⁰ employs a Graph Convolutional Network (GCN) to encode topological constraints between atoms and bonds. This method significantly enhances chemical validity without relying on explicit valence correction rules and enables the optimization of properties, such as QED and log *P*, *via* gradient ascent directly within the continuous latent space.

RNNs are generative models designed to capture sequential dependencies through recurrent architectural structures. While demonstrating efficacy in time-series domains, RNNs have been adapted for molecular generation, typically by treating molecules as sequences such as SMILES. BIMODAL leverages a bidirectional RNN architecture. By generating SMILES tokens alternately from both termini based on the parity of time steps,

BIMODAL⁹¹ resolves the non-directionality of SMILES representations and transcends the constraints of traditional unidirectional RNNs, which are limited to strict left-to-right decoding. Despite the capability of RNN-based approaches to handle variable-length SMILES representations, their performance across evaluation metrics generally underperforms other generative models. Furthermore, the training of RNN models is frequently compromised by optimization instabilities, most notably the vanishing gradient problem.

Diffusion models constitute a class of generative frameworks that generate data *via* a stepwise denoising process, by effectively learning to reverse a forward diffusion trajectory. The forward process iteratively perturbs the data with Gaussian noise governed by a predefined schedule, transforming the complex data distribution into Gaussian distribution over *T* time steps; this progression is mathematically formalized as a Markov chain. Leveraging the property that the reversal of a diffusion process is also Markovian, the generative reverse process employs a deep neural network to learn the denoising process, aiming to reconstruct the true data distribution from Gaussian noise.

Equivariant Diffusion Model (EDM)¹² represents a pioneering approach, integrating diffusion frameworks with E(3)-Equivariant Graph Neural Networks (EGNNs)⁴⁹ to *de novo* 3D molecular generation. By modeling atomic coordinates and categorical atom types as joint features within a 3D molecular graph, EDM learns to sample valid conformations directly from Gaussian noises. Evaluations on the QM9 and GEOM-drugs benchmark datasets demonstrate that EDM significantly outperforms baselines like flow-based models.

Furthermore, the GeoLDM⁹² method introduces a Latent Diffusion Model (LDM) to 3D molecular generation. It utilizes a VAE encoder to project high-dimensional molecular features into a compressed continuous latent space, where the diffusion process is executed, followed by structural reconstruction *via* a VAE decoder. This architecture addresses the challenges associated with the joint modeling of heterogeneous data types like continuous 3D coordinates and discrete atom types. By operating in a latent space, GeoLDM effectively smooths the feature embeddings, thereby enhancing training robustness.

SBDD shares methodological parallels with PLDS; however, it places a distinct emphasis on the 3D structural generation of ligands, treating target information as conditional constraints. In recent years, the integration of AI algorithms such as transformers and diffusion models, has significantly enhanced the efficiency of SBDD (Table 7).

AR methods are a class of generative models that model sequential data based on the chain rule of probability, decomposing the joint distribution of a sequence into a product of conditional probabilities. For a given sequence ($x_1, x_2 \dots x_T$), its joint probability distribution can be expressed as: $P(x_1, x_2 \dots x_T) = P(x_1) \cdot P(x_2|x_1) \cdot P(x_3|x_1, x_2) \dots P(x_T|x_1, x_2, \dots, x_{T-1})$. Consequently, sequence generation is modeled as the stepwise generation of constituent elements, where each element is conditioned on all preceding ones $x_1, x_2, \dots, x_{T-1}$. The AR models have been successfully extended to SBDD to model the probability

Table 7 Recent AI-based algorithms for structure-based drug design

Method	Model	Representation	Year	Reference
Luo <i>et al.</i>	AR	3D point cloud	2021	97
Grechishnikova <i>et al.</i>	Transformer	SMILES	2021	98
SBMolGen	RNN	SMILES	2021	100
RELATION	VAE	3D property grids	2022	101
Pocket2Mol	GNN	3D graph	2022	13
CMolGPT	Transformer	SMILES	2023	99
DiffSBDD	Diffusion	3D graph	2024	14

distributions of atomic coordinates and types within druglike molecules.

Notably, Luo *et al.*⁹⁷ proposed an algorithm for 3D molecular structure generation that applies a conditional probability density function, $p(e|r, C)$, to represent the probability of an atom of type e appearing at position r , given the binding site context C . During training, the model employs a masking strategy where a portion of the complete molecule is masked, and the algorithm predicts the missing components based on the remaining structure. The training objective involves multi-task modeling, utilizing a composite loss function that includes binary cross-entropy, multi-class cross-entropy, and boundary atom prediction loss. This ensures high accuracy in generating atomic coordinates, determining atom types, and deciding generation termination. Furthermore, Luo *et al.* innovatively incorporated a rotation- and translation-invariant GNN to encode atoms within the binding site, demonstrating outstanding performance in terms of binding affinity, molecular diversity, and drug-likeness.

Transformer is an autoregressive deep learning framework grounded in the self-attention mechanism. Originally designed for sequential data, it has subsequently been adapted for applicability in graph-structured data. By applying positional encodings to preserve sequential order and attention mechanisms to compute element-wise correlations, the transformer effectively captures data context, demonstrating robust capabilities in modeling long-range dependencies.

Transformers, especially the Generative Pre-trained Transformer (GPT), have been successfully introduced in SBDD. Through large-scale unsupervised pre-training, these models implicitly learn both SMILES syntax and latent pharmaceutical properties. Grechishnikova *et al.*⁹⁸ proposed a method that frames targeted drug design as a sequence-to-sequence translation task, that is, mapping protein amino acid sequences to ligand SMILES strings. This approach offers an end-to-end molecule generation without requiring 3D structural data; however, it overlooks the complexity of protein–ligand interactions, specifically neglecting that it is rarely a simple one-to-one mapping.

To address the target specificity, CMolGPT⁹⁹ employs a pre-training strategy on the MOSES⁶¹ dataset followed by fine-tuning on target–ligand pairs. This process incorporates target-specific constraints, yielding molecules with high quality and bioactivity compared to baseline methods. Nevertheless, these sequence-based approaches are fundamentally limited by

their inability to model the 3D spatial structural information of protein–ligand interactions, resulting in a lack of deep mechanistic insight.

The application of diffusion models in SBDD integrates robust 3D structural representations with state-of-the-art generative capabilities. DiffSBDD¹⁴ proposes a conditional diffusion model which frames SBDD as a conditional generation dependent on the binding pocket structure. Conditional diffusion models are generally categorized into classifier-guided and classifier-free variants. The latter incorporates conditional constraints explicitly during training, thereby avoiding external dependencies. In contrast, the former requires an additional classifier architecture to inject gradient-based guidance throughout the sampling process. DiffSBDD introduces two distinct classifier-free generation strategies. The first strategy employs fixed-pocket conditioning, where the protein pocket coordinates remain static. An Equivariant Graph Neural Network (EGNN) is applied to learn the noise upon the ligand's 3D structure, utilizing the ground-truth pocket as a static condition at each denoising step. The second strategy models the joint distribution of the ligand–protein complex. In this mode, the EGNN simultaneously estimates noise for both ligand and pocket, allowing the pocket structure to be iteratively updated as a dynamic context throughout the sampling process.

Currently, diffusion models represent the dominant methods in AI-driven *de novo* drug design. However, the great potential for advancement remains, particularly regarding the fusion of multi-modal molecular information and the modeling of dynamic protein–ligand structural interactions.

6 AI for drug property optimization

Following the identification of hit compounds in the traditional drug design pipeline, it is essential to improve their therapeutic potential by enhancing efficacy, safety, and pharmacological properties. Conventionally, this process is expert-driven, relying on empirical knowledge to modify functional groups or employing scaffold hopping—for instance, by replacing the core of a natural product template with that of the hit compound. The implementation of AI methods has significantly accelerated the property optimization process. In contrast to the need for explicit rules, AI methods implicitly navigate the neighboring chemical space for optimal molecular structures.

AI-driven drug property optimization typically initiates either *via de novo* generation or by modifying initial molecular

scaffolds, aiming to produce a set of molecules that satisfy desired properties. These properties must be computable or predictable from the molecular structure; they range from empirical properties such as QED and $\log P$ to complex properties predicted by trained AI classifiers, such as binding affinity scores for specific targets.

As shown in Table 8, recent AI-based methods for property optimization are mainly grounded in Reinforcement Learning (RL)¹⁰² or Markov Chain Monte Carlo (MCMC) algorithms.¹⁰³ Specifically, RL-based approaches, such as Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), and Actor-Critic (AC), model the molecular generation and optimization task as a Markov Decision Process (MDP). In a MDP $M(S, A, Pa(s, s'), Ra(s, s'))$, S represents the state space which is the current molecule structure; A represents the action space, which allows chemically valid edition on the molecule; $Pa(s, s')$ and $Ra(s, s')$ represent the probability of the transition and reward from the state s to the state s' under the action a , respectively, and the reward is set as the combinations of the molecular properties.

Furthermore, considered that there are diverse molecular properties involved, property optimization constitutes to be a multi-objective optimization task. Many algorithms address this by converting multiple objective functions into a single-objective problem *via* weighted summation.¹⁰⁴ However, this strategy overlooks the implicit correlations existing between different molecular properties. Moreover, the model performance under this strategy is highly sensitive to the hyperparameters, specifically the weights assigned to each property.

However, current computational properties like QED and SAScore do not work well for all drug molecules. To address this limitation, QADD²⁷ proposes a DQN-based RL method combining with a GNN-based molecular quality assessment model for molecule generation and property optimization. QADD defines a molecular quality score to evaluate drug potential, computed *via* a GNN trained on bioactive molecules as positive samples and low-quality molecules generated by RL models as negative samples. Additionally, QADD introduces a multi-objective RL framework for jointly optimizing multiple molecular properties, wherein a GNN classifier is trained to predict the target binding affinity property. By integrating an iterative refinement strategy, QADD yields promising performance in generating novel, high-quality drug candidates with high binding affinity to biological targets like DRD2.

MLPS¹⁷ presents an Auto Encoder (AE) model combining with local Bayesian optimization models for multi-objective molecular property optimization. The AE model transforms the discrete chemical space into a continuous latent space, while local Bayesian optimization models are applied to search for local optimal solutions. This strategy allows MLPS to handle multi-objective molecular design scenarios while learning the comprehensive Pareto set, in contrast to a limited Pareto set produced by conventional multi-objective approaches.

ChemLLM¹⁰⁵ offers an alternative perspective on molecular property optimization. Leveraging Large Language Models (LLMs), it encodes quantitative molecular properties into textual descriptions and inputs the target molecule as a string representation such as SMILES and InChI. Consequently, optimized molecular structures are obtained through an interactive dialogue. Nevertheless, the success rate of this approach has not been systematically verified by experiments. This lack of validation raises concerns regarding whether LLMs can adequately interpret diverse chemical corpora.

7 Evaluation metrics

In the past decade, a multitude of AIDD models have been developed for the generation and screening of drug candidates. To ensure a rigorous and equitable assessment of these diverse methodologies, the selection of appropriate evaluation metrics aligned with specific task objectives is significant. These metrics span a wide spectrum, including pharmacochemical properties and similarity measurements. While certain metrics possess universal applicability, others are tailored to distinct tasks like virtual screening, *de novo* generation, or property optimization.¹⁰⁹

Across these three subtasks of drug discovery, the focus of evaluation metrics varies significantly: virtual screening typically prioritizes binding affinity and ADMET properties; property optimization centers on pharmacochemical metrics, including QED, SAScore, The Logarithm of the Partition Coefficient ($\log P$), and energy; whereas *de novo* molecule generation necessitates a diverse array of metrics, ranging from 2D topological to 3D geometric metrics, tailored to specific task objectives. This section systematically classifies the prevalent evaluation metrics in AIDD into four primary categories: 2D metrics, 3D metrics, property-based metrics, and *in vitro* potency metrics. A comprehensive understanding of these

Table 8 Recent AI-based algorithms for drug property optimization

Method	Model	Representation	Year	Reference
GCPN	PPO	2D graph	2018	15
molDQN	DQN	Fingerprint	2019	16
MolGPT	Transformer	SMILES	2021	18
MARS	MCMC, GNN	2D graph	2021	106
QADD	DQN, GNN	2D graph	2023	27
FREED	AC, GCN	2D graph	2024	107
ChemLLM	LLM	Text, string representations	2024	105
MLPS	AE, local Bayesian optimization	SELFIES	2025	17
GP-MoLFormer	Transformer	SMILES	2025	108

metrics is fundamental for the effective benchmarking of AI models and for driving further methodological innovation.

7.1 2D metrics

7.1.1 Chemical validity. Chemical validity quantifies the proportion of generated molecules that satisfy fundamental chemical constraints. A molecule is considered “valid” if its constituent atoms conform to established valency rules and charge requirements. This metric serves as a primary benchmark for assessing the ability of 2D generative models to generate structurally valid compounds. In most generative frameworks, validity checks are restricted to the heavy atoms. Consequently, strict explicit bond saturation is not mandated; for instance, a tetravalent carbon atom with fewer than four explicit bonds is deemed valid, as the remaining valency is assumed to be satisfied by implicit hydrogen atoms. Chemical validity is routinely calculated using the sanitization protocols of the RDKit tools.

7.1.2 Novelty. Novelty quantifies the proportion of generated molecules that are structurally distinct from the training dataset. This metric serves as a critical indicator of a model's generalization capability, evaluating its capacity to extrapolate beyond the learned data distribution and explore novel regions of chemical space. In scenarios where the training dataset is undefined, novelty can be benchmarked by treating molecules from databases as referenced molecules. Furthermore, it can be assessed by calculating the similarity or distributional divergence between the generated molecules and the referenced molecules. For instance, novelty can be expressed as: $\text{Novelty} = 1 - \text{mean}(\text{sim}(x, y) | x \in s, y \in s')$, where x and y represent molecules from the generated set s and the referenced set s' respectively.

7.1.3 Uniqueness. Uniqueness quantifies the proportion of distinct, non-redundant structures within the generated molecules. This metric serves as a critical diagnostic tool for detecting mode collapse, a failure case where the generative model converges to a limited set of outputs, leading to the repetitive generation of identical molecular entities.

7.1.4 Diversity. Diversity quantifies the structural variation among the generated molecules. There are various ways to calculate the diversity. For instance, it can be calculated as: $1 - \sum_{x, y \in S, x \neq y} \text{sim}(x, y) / (n(n-1))$, where x and y represent different molecules in the generated set S and n represents the total number of molecules in S . The diversity metric serves as a complement to the uniqueness metric, evaluating the generative model's capacity to produce structurally distinct molecules.

7.2 3D metrics

7.2.1 Atom stability. Atom stability assesses the structural validity of 3D generated molecules by quantifying the proportion of atoms that conform to established valency and charge constraints.⁴⁸ Unlike 2D representations (e.g., SMILES or 2D graphs) where connectivity is explicit, 3D generative models frequently output discrete point clouds lacking defined

topology. Consequently, chemical bonds and their types must be inferred from interatomic Euclidean distances. Moreover, since many 3D generative approaches apply all-atom modeling, thereby precluding the reliance on implicit hydrogens for valency satisfaction, so the criteria for atom stability are inherently more stringent than those used in standard 2D validity assessments.

7.2.2 Molecule stability. Molecule stability quantifies the proportion of generated molecules that are chemically stable in their entirety. A molecule is defined as stable if and only if all of its constituent atoms strictly conform to the atom stability criteria.⁴⁸ This metric provides a global assessment of 3D generative validity, serving as a complement to the atom stability.

7.2.3 RMSD. RMSD serves as a canonical metric for quantifying the geometric deviation between a generated 3D conformation and a ground truth or reference conformation. Formally, it is calculated as:

$$\text{RMSD}(M, M') = \frac{\sqrt{\sum_i^N ((x_i - x'_i)^2 + (y_i - y'_i)^2 + (z_i - z'_i)^2)}}{N}, \text{ where}$$

(x_i, y_i, z_i) and (x'_i, y'_i, z'_i) represent the 3D coordinates of the i -th atom in the generated conformation M and the reference conformation M' , respectively. Since the raw coordinates are sensitive to spatial orientation, the Kabsch algorithm¹¹⁰ is typically employed to compute the optimal rotation matrix that aligns (x_i, y_i, z_i) to (x'_i, y'_i, z'_i) , ensuring that the calculated RMSD represents the minimal geometric distance.

RMSD is ubiquitous in tasks constrained by a fixed conformation, such as 2D-to-3D conformation generation and molecular geometry optimization, where the model performance is evaluated from the mean RMSD. However, in most *de novo* generation tasks which aim to learn and sample from the distribution of the training set, the absence of reference structures renders RMSD calculation inapplicable.

7.3 Property metrics

7.3.1 Molecular weight (MW). MW stands as one of the most straightforward metrics for molecular evaluation and serves as a fundamental component of numerous statistics-based metrics. Quantifying the distributional divergence of MW between generated and training sets provides a foundational perspective for assessing the performance of generative models. However, MW is rarely employed as an objective function for property optimization, as there is no direct correlation between molecular weight and drug-likeness.

7.3.2 Log P . Log P ¹¹¹ is defined as the base-10 logarithm of the equilibrium concentration ratio of a solute distributed between two immiscible phases: n -octanol and water. Mathematically, it is formulated as: $\log P = \log_{10}(C_1/C_2)$, where C_1 and C_2 represent the solute concentrations in the organic and aqueous phases, respectively. Fundamentally, this metric quantifies the hydrophobicity-lipophilicity balance of a molecule. Similar to molecular weight, log P constitutes a foundational physicochemical descriptor and rarely applied as an optimization objective. Nevertheless, log P remains a critical

constraint in drug discovery, as it plays a significant role in a molecule's transport capabilities as well as the selection of synthetic routes.

7.3.3 QED. QED²⁰ is a metric designed to assess the pharmaceutical potential of a molecule. It is derived from the calculation of eight physicochemical properties: molecular weight, $\log P$, topological polar surface area, the number of hydrogen bond donors and acceptors, the number of rotatable bonds, aromatic rings, and undesirable structural alerts. These properties are integrated *via* a weighted average and normalized to the [0, 1] interval to yield the QED score; a higher score implies a greater likelihood of the molecule being drug-like. QED is frequently utilized in conjunction with other metrics to provide a systematic evaluation of drug potential and often serves as an objective function for property optimization. It can be computed using the RDKit toolkit.¹¹²

7.3.4 SAScore. SAScore²¹ is a metric designed to evaluate the synthesis difficulty of a molecule, derived from an integration of structural analysis and synthetic accessibility. The calculation process typically involves three stages: first, assessing molecular complexity based on features such as ring counts, molecular weight, and chirality; second, identifying 'difficult' motifs by scoring substructures against a reference fragment database; and finally, fragmenting the molecule to evaluate the accessibility of synthetic pathways based on standard reactions. Collectively, these components define the SAScore. The score is standardly normalized to a range of [1, 10], where a higher value indicates greater synthetic difficulty. In some studies, the scale is normalized to the [0, 1] interval. SAScore can be computed using the RDKit toolkit.

7.3.5 Lipinski's rule of five (Ro5) and Veber's rules. Ro5 (ref. 113) constitutes a set of empirical guidelines proposed by medicinal chemist Christopher Lipinski for the screening of drug-like molecules. These rules are derived from expert experience and define the following five criteria: (1) the molecular weight is less than 500 daltons; (2) the number of hydrogen bond donors including hydroxyl and amino groups does not exceed 5; (3) the number of hydrogen bond acceptors does not exceed 10; and (4) $\log P$ is less than 5. Furthermore, to better evaluate oral bioavailability, Veber *et al.*¹¹⁴ extended these guidelines by proposing Veber's rules, which state that drug-like molecules should have a number of rotatable bonds that does not exceed 10 and a polar surface area (PSA) equal to or less than 140 Å². Consequently, the combination of Ro5 and Veber's rules frequently serves as a standard benchmark for assessing the pharmaceutical potential of molecules.

7.3.6 ADMET. ADMET serves as a comprehensive set of evaluation metrics covering five critical biological processes: absorption, distribution, metabolism, excretion, and toxicity.¹⁹ These processes collectively determine the safety profile of a drug within the biological system. Incorporating ADMET evaluation in the early stages of drug design like hit identification enables the exclusion of candidates with poor safety characteristics, thereby effectively reducing time costs and material consumption.¹¹⁵

ADMET consists of over thirty specific metrics, including intestinal absorption rate, cellular permeability, half-life,

CYP450 inhibition, carcinogenicity and others. While these properties are traditionally determined *via in vitro* (cell) or *in vivo* (animal) assays, the advent of AI-driven algorithms¹¹⁶ has enabled the rapid and accurate prediction of ADMET properties, making their integration as evaluation metrics in computational workflows.

7.3.7 Energy-based metrics. The QM9 dataset⁴⁴ not only incorporates 3D conformational information but also provides relevant energy properties calculated *via* DFT. Key properties include ϵ_{HOMO} , ϵ_{LUMO} , and ϵ_{gap} . Here, ϵ_{HOMO} and ϵ_{LUMO} represent the energies of the highest occupied molecular orbital and the lowest unoccupied molecular orbital, respectively, while ϵ_{gap} represents the difference between them. Other energy properties include zero-point vibrational energy (ZPVE), internal energy (U), enthalpy (H), Gibbs free energy (G), heat capacity at constant volume (c_v) and others.

These properties reflect the molecule's optoelectronic and thermodynamic characteristics, which influence the molecular structural stability and chemical reactivity. Consequently, they are frequently employed in 3D molecular generation and property optimization tasks to evaluate whether the generated or optimized molecules have the desired properties.

7.4 *In vitro* potency metrics

7.4.1 Half maximal inhibitory concentration (IC50) and half maximal effective concentration (EC50). IC50 and EC50 are metrics designed to evaluate the *in vitro* biological potency of a molecule, derived from experimental cellular and biochemical assays.¹¹⁷ The evaluation process typically involves three stages: first, establishing a standardized cell-based or cell-free biological system; second, administering the candidate molecule across a gradient of concentrations; and finally, measuring the specific concentration required to inhibit a target biological process by half (IC50) or to induce a half-maximal biological response (EC50). Collectively, these metrics define the fundamental functional potency of a drug candidate. The values are standardly expressed in molar concentrations (*e.g.*, μM), where a lower value indicates higher potency and stronger binding affinity to the target protein.

7.4.2 Cellular Target Engagement. Cellular target engagement is a metric designed to evaluate the physical binding of a molecule to the protein target within the complex environment of a living cell. The metrics is commonly quantified through thermal shift curves or fluorescence ratios by standardized empirical platforms like the Cellular Thermal Shift Assay (CETSA),¹¹⁸ where a robust, dose-dependent signal indicates successful and specific intracellular binding.

8 Discussion

AI methods have achieved remarkable success in drug design. The recent applications include designing novel and diverse molecules, generating or screening ligands that bind to biological targets, and optimizing multiple molecular properties. Recent advancements in AI algorithms, such as diffusion models and E(3)-equivariant networks, have empowered the

field of AIDD with new capabilities. This is particularly evident in generative tasks for 3D molecular structures, where these innovations have significantly improved both the chemical validity and diversity of generated structures.

Notably, large foundational models have made significant strides in AIDD, driven by their exceptional representation learning capabilities. Leveraging self-supervised learning on billion-scale datasets, these models capture intrinsic biological and chemical rules encoded within embeddings without relying on extensive labeled data. The typical applications include protein language models^{119–121} and molecular representation learning models.^{122–124} Protein language models such as ESM3 (ref. 119) integrate sequence, structure, and function within a unified generative framework, enabling binding pocket identification and binding conformation generation by co-reasoning across these three modalities. Molecular representation learning models like Uni-Mol¹²² and GIT-Mol¹²³ have demonstrated outstanding performance in generating valid molecular structures and predicting protein–ligand interactions or molecular properties. When faced with novel data distributions across different AIDD tasks, these transferable embeddings serve as highly robust structural and physico-chemical descriptors and be applied through fine-tuning strategies. The optimal choice of fine-tuning strategies and benchmarks depends heavily on the volume of downstream tasks. For severe data-scarce tasks, utilizing frozen pretrained embeddings coupled with a lightweight deep learning top is increasingly preferred to prevent overfitting and preserve the learnt features of the pretraining data. Conversely, when the training data are sufficient, full end-to-end fine-tuning or parameter-efficient fine-tuning techniques, such as Low-Rank Adaptation (LoRA), can be deployed to align the generalized embeddings with task-specific distributions. By effectively transferring knowledge from vast unsupervised data to downstream tasks, these pretrained foundational models offer a promising way to address the limitations of traditional task-specific approaches.

However, existing approaches still exhibit certain limitations, which continue to raise doubts regarding the reliability of AI-driven approaches.

Primarily, the AI methods are constrained by data scarcity, consequently exhibiting limited generalization capabilities. Despite the updating of molecular databases like ChEMBL²³ and PDBbind,⁶⁰ high-quality, 3D labeled data for protein–ligand complexes remain scarce. This data sparsity creates a significant bottleneck, particularly for PLIP and SBDD tasks. Models trained on these limited data often struggle with out-of-distribution (OOD) generalization, failing to accurately predict interactions for novel protein targets or biomolecules with distinct scaffolds.¹²⁵ The long-tail distribution of bioactivity data further exacerbates this,¹²⁶ biasing models toward well-studied targets while neglecting orphan targets. Although transfer learning and pre-training strategies are designed to help alleviate this problem, the resulting generalization capabilities are still not satisfactory.

Moreover, some AIDD approaches suffer from potential information leakage issues, such as those arising from the

failure to partition training and test sets based on molecular scaffolds. Given the high degree of structural similarity typically inherent in small-molecule datasets, random splitting inevitably introduces information leakage, thereby yielding over-optimistic evaluation metrics. A rational mitigation strategy is to partition the dataset according to compound scaffolds and substructures, which rigorously evaluates the model's predictive capability in unexplored chemical spaces. Similarly, network-based prediction models are also prone to potential information leakage. For instance, when all ligands and receptors are integrated into a single global network, utilizing a subset of nodes and their associated edges for training while reserving others for testing often exposes the test nodes within the training process. To mitigate this, a cold-start splitting strategy can be applied, where a subset of drug or protein nodes is entirely excluded from the training graph to evaluate the model's capability on completely unseen targets. Furthermore, employing degree-preserving negative sampling forces the model to learn the biophysical interactions rather than merely memorizing node degrees or network connections.²⁵

Additionally, the lack of robust metrics especially in 3D molecule generation and docking simulation tasks limits the performance evaluation and real-world application of AI models. The validity, atom stability and molecule stability serve as metrics for 3D molecule generation, while these metrics can evaluate neither the reasonability of generated conformations or the synthetic accessibility. The diversity and novelty for 3D molecular conformations are not been defined. These limitations underscore the need for a comprehensive metrics for 3D molecule generation. Moreover, chirality is frequently ignored while generating 3D molecules owing to the popular E(3) equivariant networks,¹²⁷ and also tend to be ignored when using 2D graphs as molecular representations.

Furthermore, a central debate in current AI-based drug design, particularly in SBDD revolves around the trade-off between modeling resolution and computational efficiency. As discussed in previous sections, many existing SBDD methods employ coarse-grained representations⁶ (e.g., modeling residues as single nodes or C α) to reduce computational complexity and memory footprint. While efficient, this strategy inevitably oversimplifies the protein topology. By discarding side-chain atomic details, coarse-grained models often fail to capture critical non-covalent interactions and may lead to physically invalid generated structures characterized by intermolecular clashes. Conversely, surface-based representations like SurfDock⁷ and all-atom modeling like AlphaFold 3 (ref. 10) and RoseTTAFold All-Atom⁴³ offer higher fidelity. All-atom models, while computationally expensive, are essential for understanding the precise induced-fit effects and dynamic conformational changes in protein–ligand binding.¹²⁸ Despite the powerful diffusion algorithms, yet achieving high precision for protein–ligand complexes in general all-atom models remains a challenge that requires further optimization. A potential future direction could be balancing these approaches by employing coarse-grained models for rapid search space pruning and all-atom models for final high-precision refinement.

Lastly, a critical gap persists between computational predictions and biological efficacies. While current AI methods can efficiently optimize empirical computational metrics like QED and SAScore, a high score in these metrics does not guarantee real-world cellular activity or synthesizability. *In vitro* cellular models and assays provide the indispensable first layer of biological validation, serving as the experimental ground truth for AI-driven predictions. These cell-based systems not only benchmark the predictive performance of models against actual biological responses but also bridge the gap between *in silico* design and real drug discovery workflows. Consequently, a vital future direction could be establishing closed-loop workflows¹²⁹ where AI predictions are rapidly validated by automated wet labs, creating a feedback loop that iteratively refines the AI models and enhances the applicability.

9 Conclusions

AI-based approaches especially deep learning-based approaches have been applied in drug design with remarkable performance. This review introduces an overview of the integration of AI approaches in drug design, and mainly focused on three subtasks including virtual screening, *de novo* molecule generation, and property optimization. We expect that this review will provide a useful guide for future AIDD studies.

Author contributions

Conceptualization, X. P. and H.-B. S.; methodology, Y. F.; writing – original draft preparation, Y. F.; writing—review and editing, X. P. and H.-B. S. All authors have read and agreed to the published version of the manuscript.

Conflicts of interest

The authors declare no conflicts of interest.

Data availability

No primary research results, software or code have been included and no new data were generated or analysed as part of this review.

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